

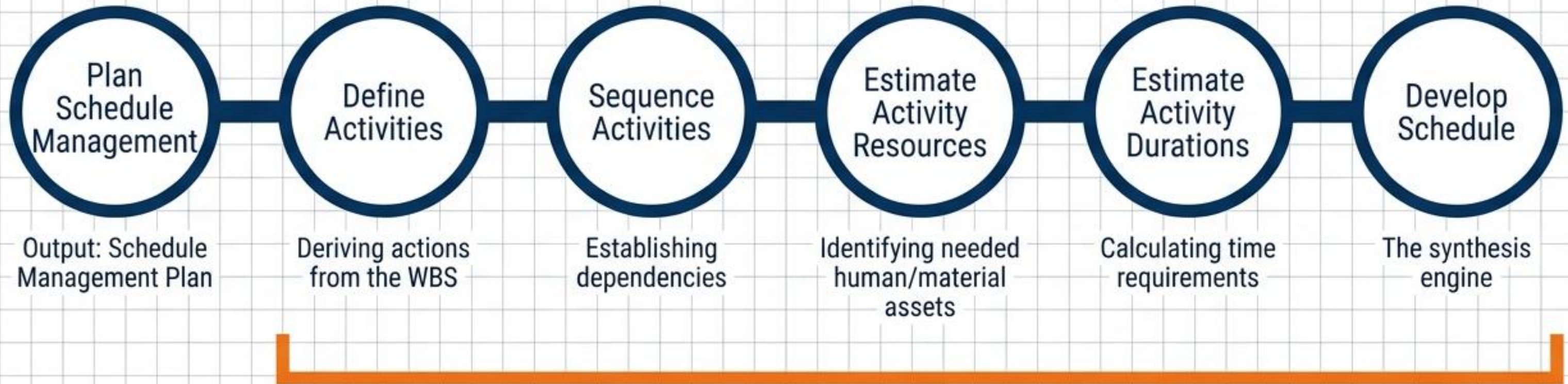
Certyfikat PMP: Project scheduling - summary

Mastering Time Management Processes,
Compression, and Baselines



dr Zbigniew Galar

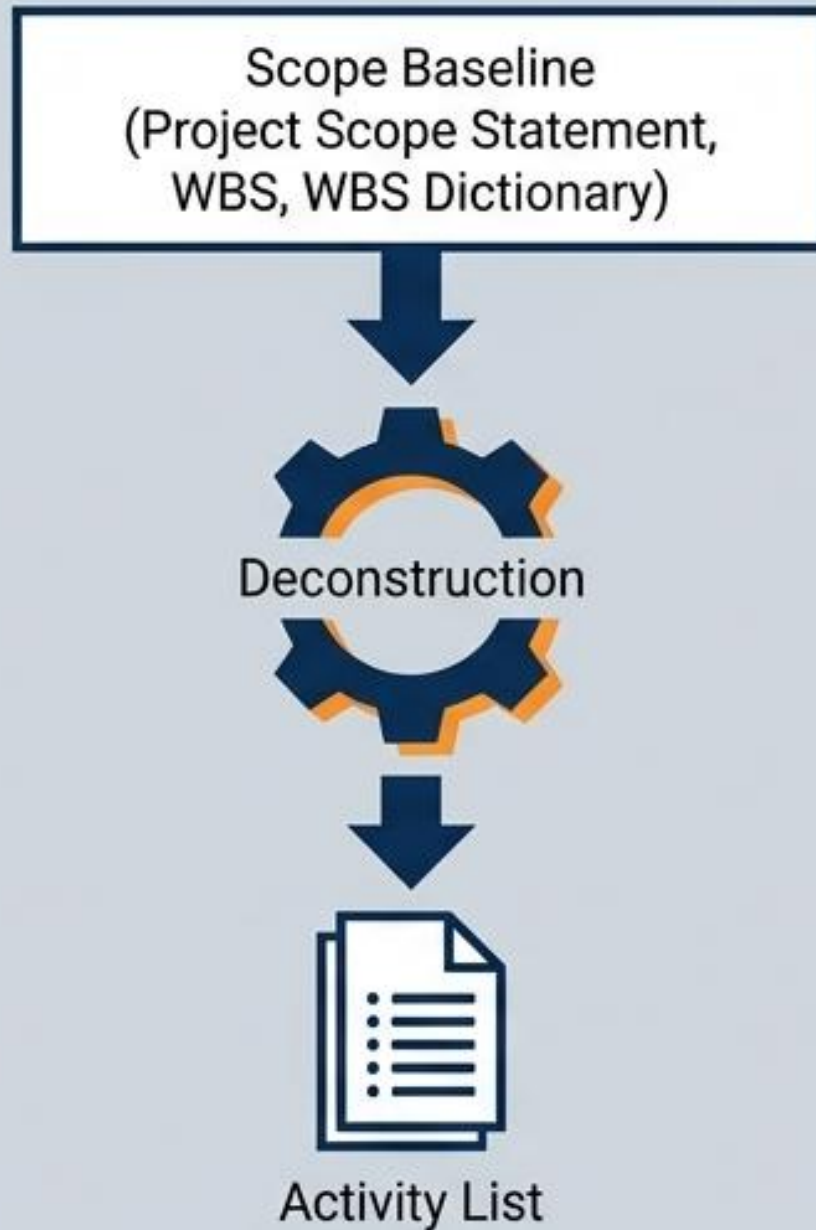
The Six-Step Scheduling Architecture



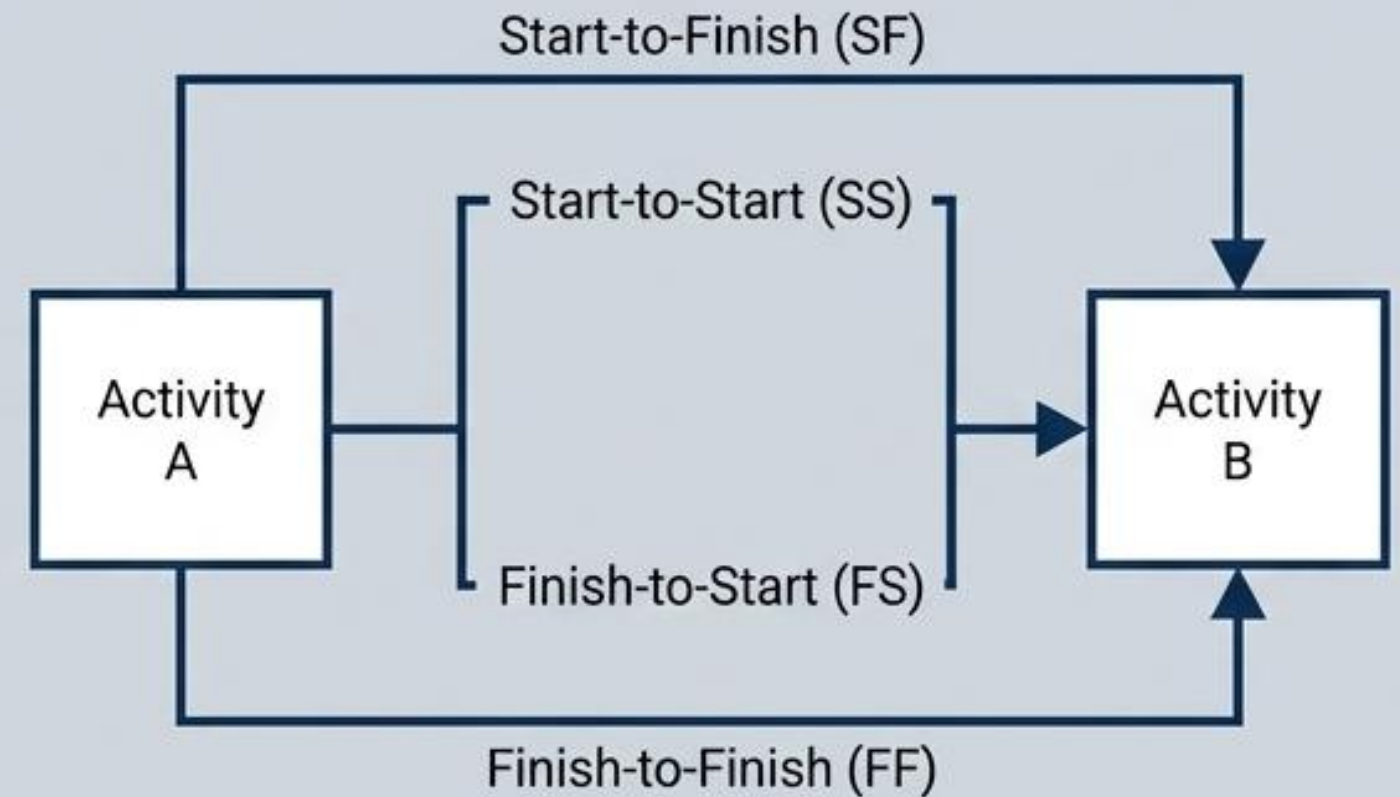
Iterative Execution: Often completed concurrently on small-to-medium projects.

From Scope Definitions to Sequential Logic

Define Activities



Sequence Activities



Dependencies dictate flow: Mandatory (Hard logic), Discretionary (Preferred logic), External, or Internal.

Establishing Duration and Resource Estimates

	Mechanism	Accuracy	Primary Use Case
Analogous (Top-Down)	Uses expert judgment and historical information.	Lowest	Early phases when detailed data is scarce.
Parametric	Multiplies a known unit by historical time per unit.	Medium-High	Quantifiable, repetitive tasks (e.g., running cable).
Three-Point (PERT)	Weighted average accounting for risk and uncertainty.	Highest	Complex projects with high uncertainty or new scope.

Calculating Confidence with PERT Estimates

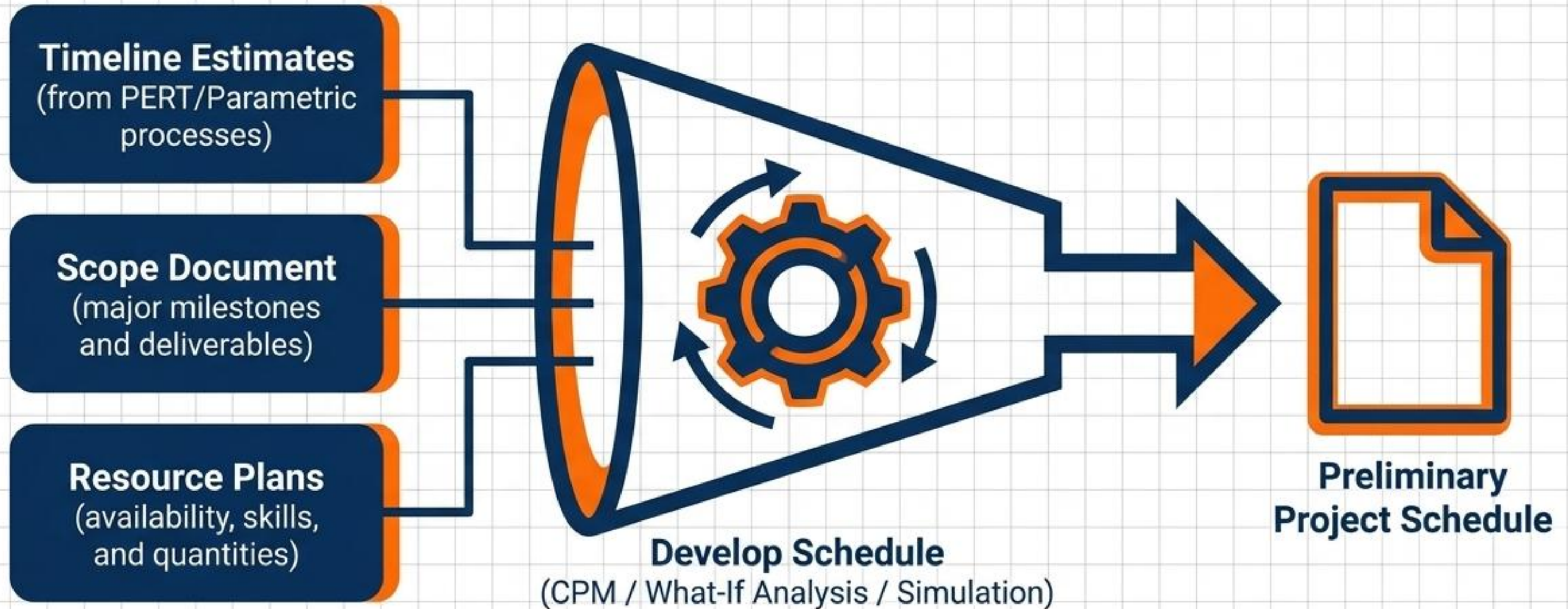


Confidence Intervals

- ± 1 Standard Deviation = 68.26% completion probability.
- ± 2 Standard Deviations = 95.44% completion probability.

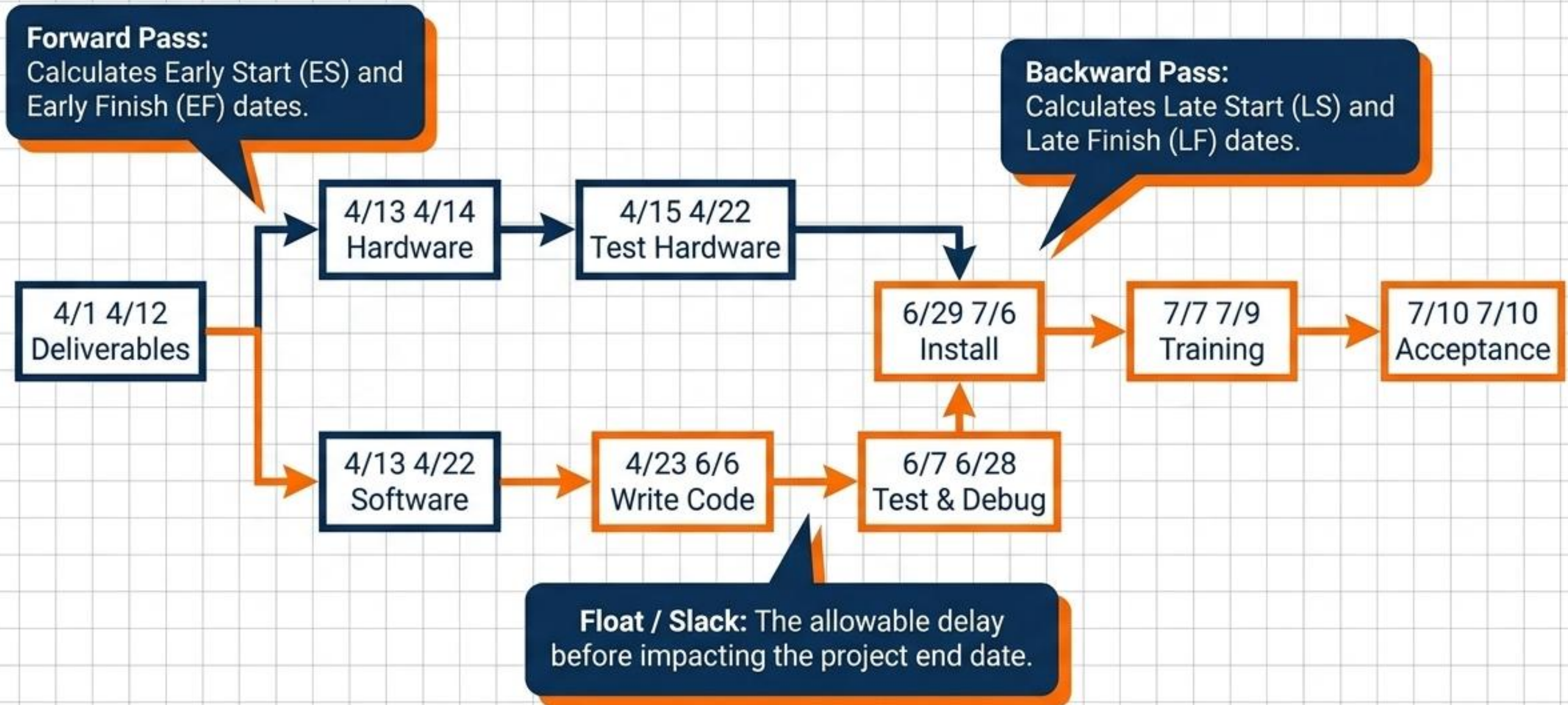
$$(O + P + (4 \times M)) / 6 = \text{Expected Duration}$$

The Schedule Development Engine



PMP Rule: The schedule model is strictly preliminary until human and material resources are formally assigned and committed.

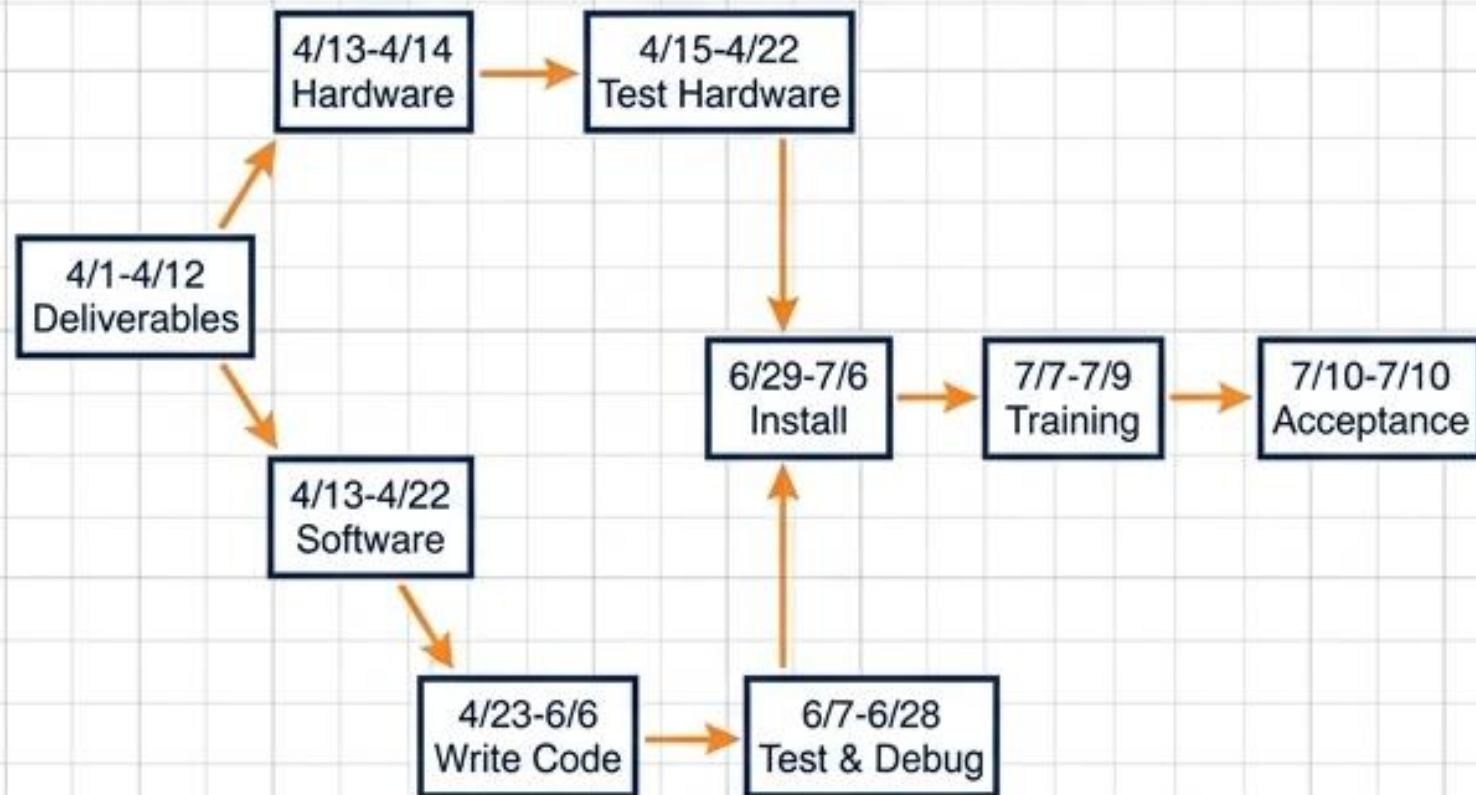
Tracing the Critical Path



Critical Path = The longest path through the network schedule = Zero Float.

Two Presentation Lenses for Schedule Management

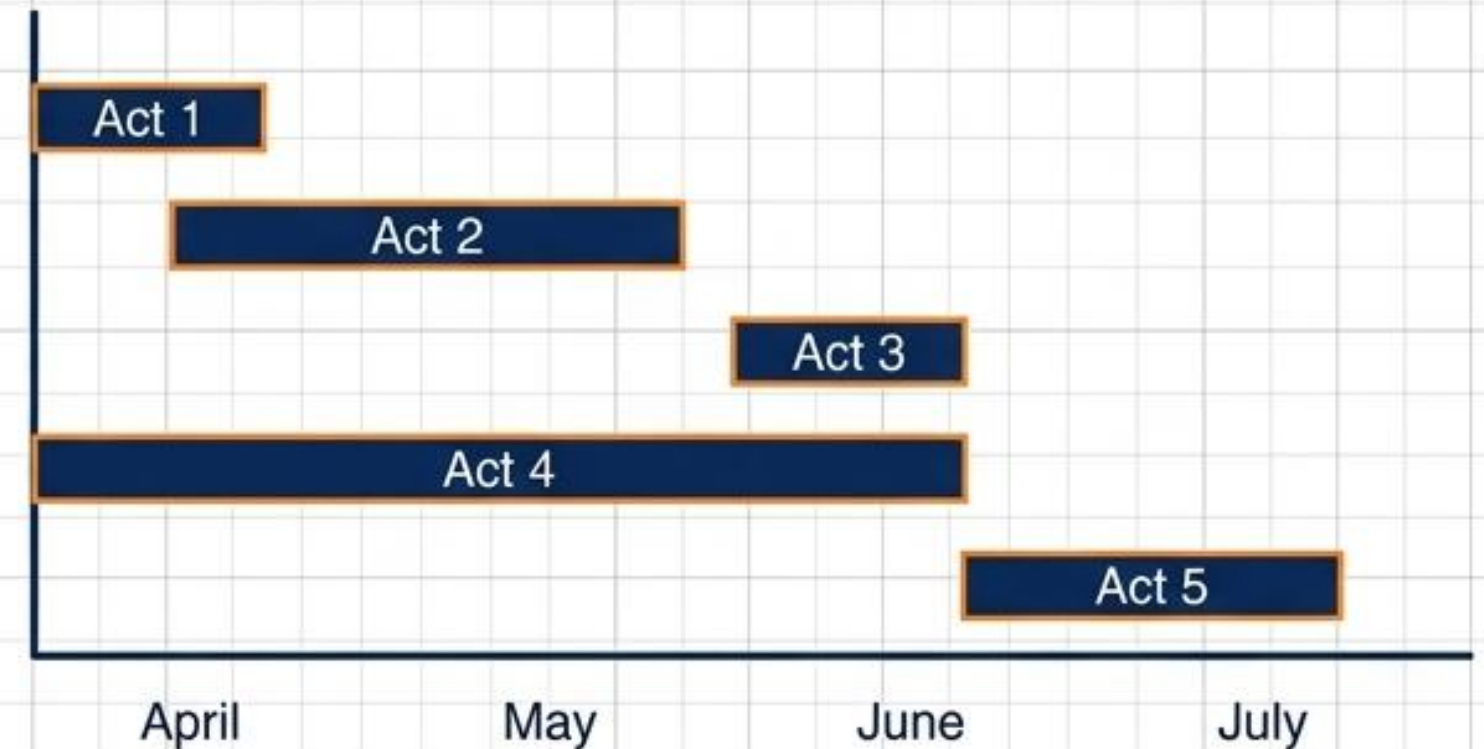
The Logic Lens



Diagnostic

Focus: Activity dependencies, sequential flow, and the Critical Path.
Audience: Project Managers and the execution team.

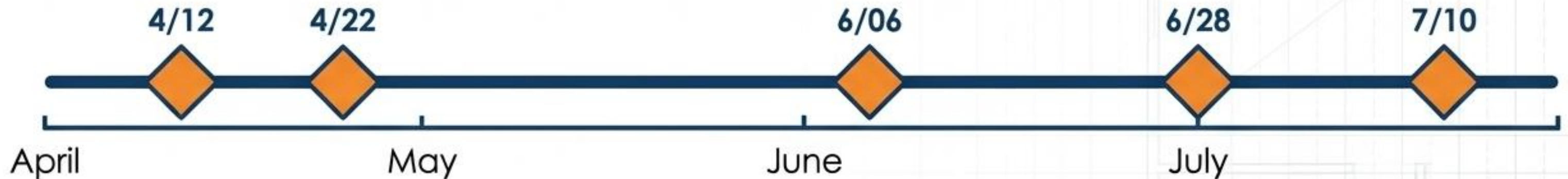
The Timeline Lens



Diagnostic

Focus: Duration, activity start/end dates, and resource overlaps against a calendar timeline.
Audience: Mid-level management tracking execution over time.

Milestone Tracking for Executive Leadership



Milestone	Scheduled Date	Actual Date
Sign-off on deliverables	4/12	4/12
Sign-off on hardware test	4/22	4/25
Programming completed	6/06	(Blank)
Testing completed	6/28	(Blank)
Acceptance and sign-off / Project closeout	7/10	(Blank)

Takeaway: Milestone charts strip away activity complexity to present a high-level view of major deliverables—the ideal presentation format for executive teams and sponsors.

Schedule Compression: Strategies for Fixing Delays



Crashing

Mechanism: Adding resources or approving overtime for Critical Path tasks.

Trade-off: Time vs. Cost.

Case Study Reality: Paying a specialized contractor \$10,000 to run Ethernet cable on an expedited timeline.



Fast Tracking

Mechanism: Performing phases or tasks in parallel that were originally scheduled sequentially.

Trade-off: Time vs. Risk.

Warning: Does not change the critical path, but significantly increases the risk of rework.

Resource Optimization: Resolving Bottlenecks

Resource Leveling

Scenario: A key resource is overallocated and faces burnout.

Mechanism: Activities are rearranged or delayed to resolve the overallocation.

 **Critical Impact:** Will likely extend the project end date and alter the Critical Path.

Resource Smoothing

Mechanism: Modifies activities strictly within the limits of their Total and Free Float.

✓ **Critical Impact:** Protects the baseline project end date and maintains the existing Critical Path. Limits optimization options.

Locking the Schedule Baseline

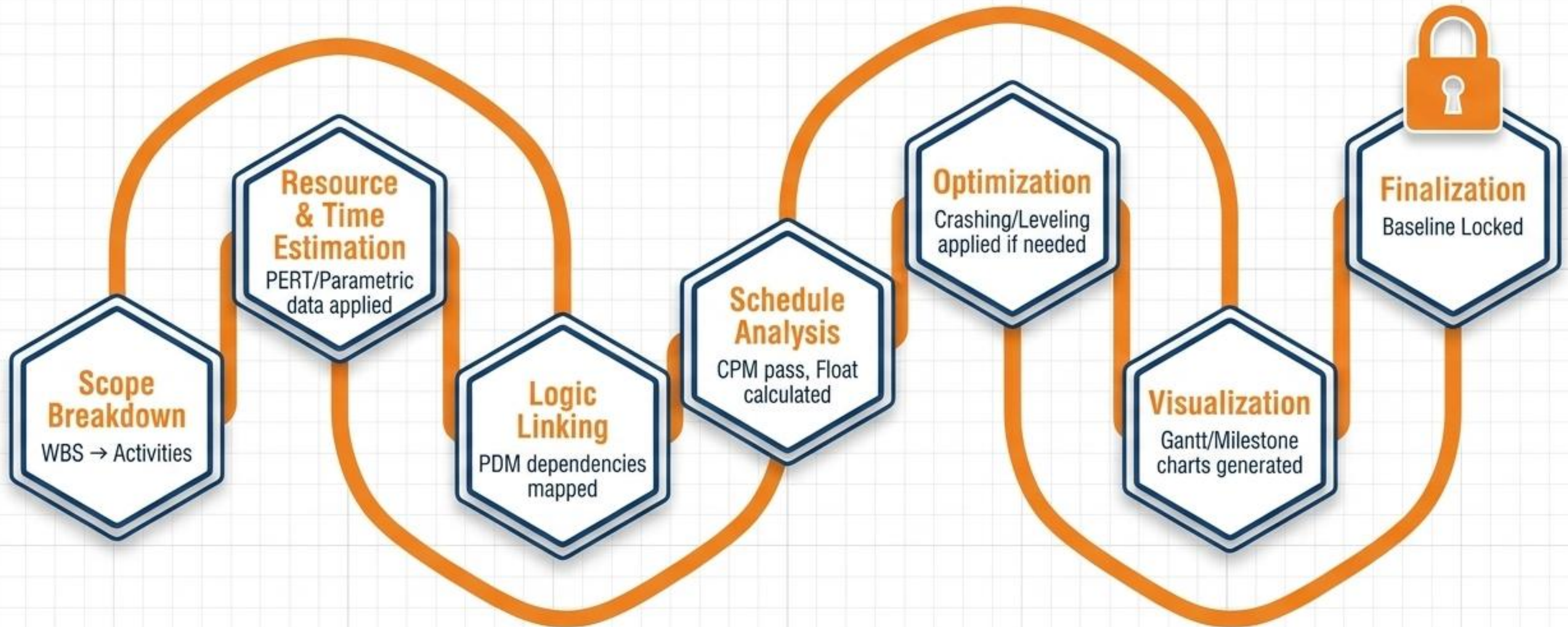


- Serves as the definitive metric against which all project execution is monitored and tracked.
- Integrated directly into the overarching Project Management Plan.
- Once locked, it can only be altered through formal Integrated Change Control procedures.

Documenting the Schedule Ecosystem

Schedule Data (Supporting Documentation)	Project Document Updates (Living Documents Altered)
• Milestones and activity attributes. • Assumptions and schedule constraints. • Resource histograms (displaying hours needed vs. time periods). • Contingency schedule reserves.	• Activity resource requirements. • Resource and project calendars. • The Risk Register (updated with new schedule-based risks).

End-to-End Schedule Synthesis



Critical Takeaways for the PMP Exam

- ✓ **The Critical Path:** It is always the longest path in the network diagram and features zero or negative float.
- ✓ **Preliminary Status:** A project schedule is only preliminary until resources are assigned and committed.
- ✓ **Leveling vs. Smoothing:** Resource leveling can change the critical path/end date; resource smoothing cannot.
- ✓ **Compression Rules:** Crashing increases cost (adds resources); Fast tracking increases risk (parallel tasks).
- ✓ **PERT Calculation:** Must know the Beta Distribution formula:
$$(\text{Optimistic} + \text{Pessimistic} + (4 \times \text{Most Likely})) / 6.$$